Which Component Carries the Effect? Ablating an ITS-Inspired Generation Framework

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Design Science Research: we design and evaluate a reusable framework for generating curriculum-constrained learning material at class and learner level. Inspired by Intelligent Tutoring System (ITS) component models, it separates a **curriculum model**, a **lesson model**, an **expert model** and a **learner model**, and combines them through a generation-and-validation process. We instantiate it on Nigerian junior-secondary (JSS1) mathematics and add one component at a time. We are aware of no published component-additive ablation of an educational LLM generation system; it lets us ask which component carries the effect.

1 Framework, instantiation, evaluation

The framework's intended applicability is generic; the evidence below is specific to the evaluated instantiation.

Each rung adds one component: **C0** curriculum model, objectives only; **C1** + lesson model, the teacher's lesson plan and its instructional specification; **C2** + expert model, scoped retrieval over her lesson note, the subject knowledge that realises the plan; **C3** + a validation-and-repair agent, a processing component, not a further knowledge model. All rungs share one generator, artefact spec and output contract, so a rung's effect is the component's, not a change of system. A second part adds the learner model: **P1** personalises frozen class-level material (**P0**) from a pupil's marked script — score, solution steps, errors, misconceptions.

2 Evidence from the prototype

Nigerian JSS1 mathematics: forty pupils, one teacher. Six lessons $\times$ 4 conditions $\times$ 3 draws $= 72$ packs over 17 objectives, at temperature 0 with caching disabled. Primary: the lesson-plan-based family (C1–C3) against the objectives-only baseline.

A blinded LLM judge, run twice independently, is repeat-stable (movement 0.076; 92.4% of 288 scores identical). The family beats C0 by $+0.338$, positive in all six lessons (Wilcoxon $p = .031$). The component diagnostics locate it: **the lesson model carries all of it** — C1$-$C0 $= +0.368$ (6/6 lessons, $p = .031$), while C2$-$C1 $= -0.042$ (2/6) and C3$-$C2 $= -0.007$ (1/6). Sharpest as a floor: 8 of 18 C0 packs carry a judged score below 4, against **0 of 18** at each of C1, C2 and C3. Component execution is verified without a rater: source refs on 168 C2 and 157 C3 blocks against none at C0/C1, and a validator logging 4.22 findings per pack that closed its loop on only 6 of 18.

Four practising mathematics teachers, blinded and counterbalanced (ANOVA on position $F = 0.01$, $p = .998$), returned 1,199 of 1,224 scores. On them every contrast is **null**, including the pre-registered C1$-$C0 $= +0.044$ (95% CI $[-0.195, +0.283]$). That panel could not have resolved the contrast either way: rater identity carries 25.9% of score variance against the packet's 14.6%, $\mathrm{ICC}(1,1) = -0.163$, 91% of scores are a 4 or a 5, and the minimum detectable effect ($\approx$0.34) exceeds the largest present (0.152). We report both instruments, not the one that separates.

Contributions. (i) A reusable ITS-inspired framework separating curriculum, lesson, expert and learner models behind one generation-and-validation process, with the component-additive ablation that makes each separately measurable; (ii) a prototype built from artefacts a Nigerian school already produces, with corpus and blinded protocol; and (iii) the observation that none of COGENT [1], FACET [2] or LectūraAgents [3] reports inter-rater reliability or validates its judge — a property the field assumes. The learner-model rung is instantiated and under evaluation.

References. [1] Liu et al., *BEA/ACL*, 2025. [2] Gonnermann-Müller et al., *arXiv:2508.11401*. [3] Sesay et al., *arXiv:2606.16428*.

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